

KEDAR DAMALE

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Portfolio | GitHub | Kaggle | LinkedIn

SUMMARY

Machine Learning Engineer with a strong mathematical foundation in linear algebra, calculus, probability, and statistics, applied to Machine Learning, Deep Learning, and Generative AI. Proficient in Python, SQL, PyTorch, and Scikit-learn, with hands-on experience building agentic AI architectures, RAG pipelines, and LLM applications using LangChain, LangGraph, and the OpenAI API. Skilled in full-stack AI/ML development and MLOps with FastAPI, Docker, and PostgreSQL, delivering systems that cut manual processing time by over 99%.

SKILLS

- **Foundational Mathematics** : Linear algebra, Calculus, Probability and Statistics
- **Programming** : Python, SQL, C, DSA, Bash
- **Data Science** : NumPy, Pandas, Polars, Matplotlib, Seaborn, Scikit-learn, Exploratory Data Analysis
- **ML & Deep Learning** : Supervised & unsupervised learning, regression, classification, clustering (DBSCAN), XGBoost, PyTorch, CNNs, YOLOv8, model evaluation, hyperparameter tuning
- **Generative AI** : LangChain, LangGraph, OpenAI API, agentic architectures, RAG, HITL design
- **Data Engineering & MLOps** : PostgreSQL, MySQL, MongoDB, DuckDB, ETL pipelines, Celery, Redis, Docker, FastAPI
- **Soft Skills** : Leadership, Requirement analysis, Team collaboration and ability to explain

WORK EXPERIENCE AND INTERSHIPS

ML and GenAI Developer

Globalspace Technologies Ltd.

Jan 2026 – Present

Navi Mumbai, MH, India

Budget Automation & Incentive Calculator | GitHub

- The legacy platform maintained a separate execution model per MR, workable only at demo scale (~20–30 employees) but unmanageable across a real hierarchy of thousands, so designed and built a dynamic scoring engine that auto-allocates MRs into the hierarchy tree as they join or leave, replacing per-MR trees with one scalable model.
- Added an agent layer with human-in-the-loop (HITL) validation to the incentive calculator, letting business users configure incentive and target-allocation policies without the deep schema/admin expertise SAP and Salesforce incentive tools require, and built a revision system to rebalance allocations after manager edits.

PatGPT | GitHub

- Built PatGPT, an agentic sales-analytics assistant over 3M+ pharmaceutical sales and tender records, driving the shift from a RAG-based approach to a planning agent that answers decision-making questions via forecasting, simulation, prediction, and historical analysis with HITL checkpoints and evidence-bound explanations.
- Owned backend architecture and DevOps for PatGPT: an asynchronous run API with SSE streaming and idempotent retries, plus a checksum-verified data promotion/rollback pipeline for safely shipping new snapshots to production.

Automation Developer (Freelance) | Link

Propelligence Advisors Pvt. Ltd.

May 2025 – Jan 2026

Remote | Thane, MH, India

- Manual PR vs GSTR-2B reconciliation of 10,000+ invoice records required 5–6 days per cycle due to non-exact invoice matching and human validation, so developed and engineered an automated PR–GSTR-2B reconciliation pipeline applying fuzzy similarity algorithms ($\geq 80\%$ match threshold) to normalize invoice fields, cutting reconciliation time by >99.9%.
- Partnered with Propelligence to build Granska (| GitHub), an audit automation suite now at 5 production modules, adopted by 2 CA firms and used by ~20 auditors, engineered for performance with Polars, RapidFuzz, and constant-memory report generation.

PROJECTS

Chessablanka – Chess Position Analyzer | GitHub

Apr 2025 – Jan 2026

- Evaluating a chess position from a photo normally needs an expert, so built an end-to-end CV pipeline with board detection, grid mapping, and a custom YOLOv8 model trained on 700+ images (augmented to 1,700 via Roboflow).
- Achieved 98.57% detection accuracy on 500+ test images and piped positions into Stockfish for instant real-time evaluation and best-move analysis.

StudyONE – Student Productivity & Collaboration Platform | GitHub

Aug 2024 – Oct 2024

- Students relied on scattered tools for notes and collaboration, so built a full-stack MERN platform with JWT auth, a personalized dashboard, and a real-time collaborative whiteboard.
- Manually converting lectures into notes was slow, so integrated Rev.ai and Google Cloud Speech-to-Text into an AI notes pipeline and deployed the app to production on Render.

Cattle Monitoring using DBSCAN and ESP32 | GitHub

Oct 2025 – Nov 2025

- Farmers had no way to track free-grazing cattle without manual rounds, so led a team to build a distributed IoT system (1 ESP32 master + N child nodes) streaming GPS/RSSI data to a Flask/MongoDB backend, applying DBSCAN clustering to auto-identify grazing zones on a Next.js live-tracking dashboard.

NeuroTRIBE-HBN – Cortical Response Analysis for ADHD Research | GitHub

Aug 2026 – Present

- No existing tool tests whether ADHD alters stimulus-evoked cortical response against an objective baseline, so built a research pipeline applying TRIBE v2 to Healthy Brain Network movie-fMRI data, comparing confirmed-ADHD participants against a matched cohort via an autonomous, Docker-Compose orchestration layer that treats legally-gated inputs as non-blocking.

EDUCATION

Bachelor of Engineering in Information Technology

Phoenix Academy of Management and Technology, affiliated to University of Mumbai

Mumbai, MH, India

- CGPA: 8.5/10

ACHIEVEMENTS

- **Qualified GATE** — Data Science and Artificial Intelligence (DA) paper
- **Published Automation Suite for CA & Audit Firms** — Adopted by 2 CA firms and used by about 20 auditors
- **First Place, College Hackathon** — Secured 1st position at college-level hackathons